

PRESS RECORD

COMPX₃₄₁ - Advanced Software Engineering Design



THE UNIVERSITY OF
WAIKATO
Te Whare Wānanga o Waikato

Software Construction Tools & Example

COMPX₃₄₁

Overview

- Software Construction Tools
- Example Product Backlog & Sprints

Software Construction Tools

Software Construction Tools

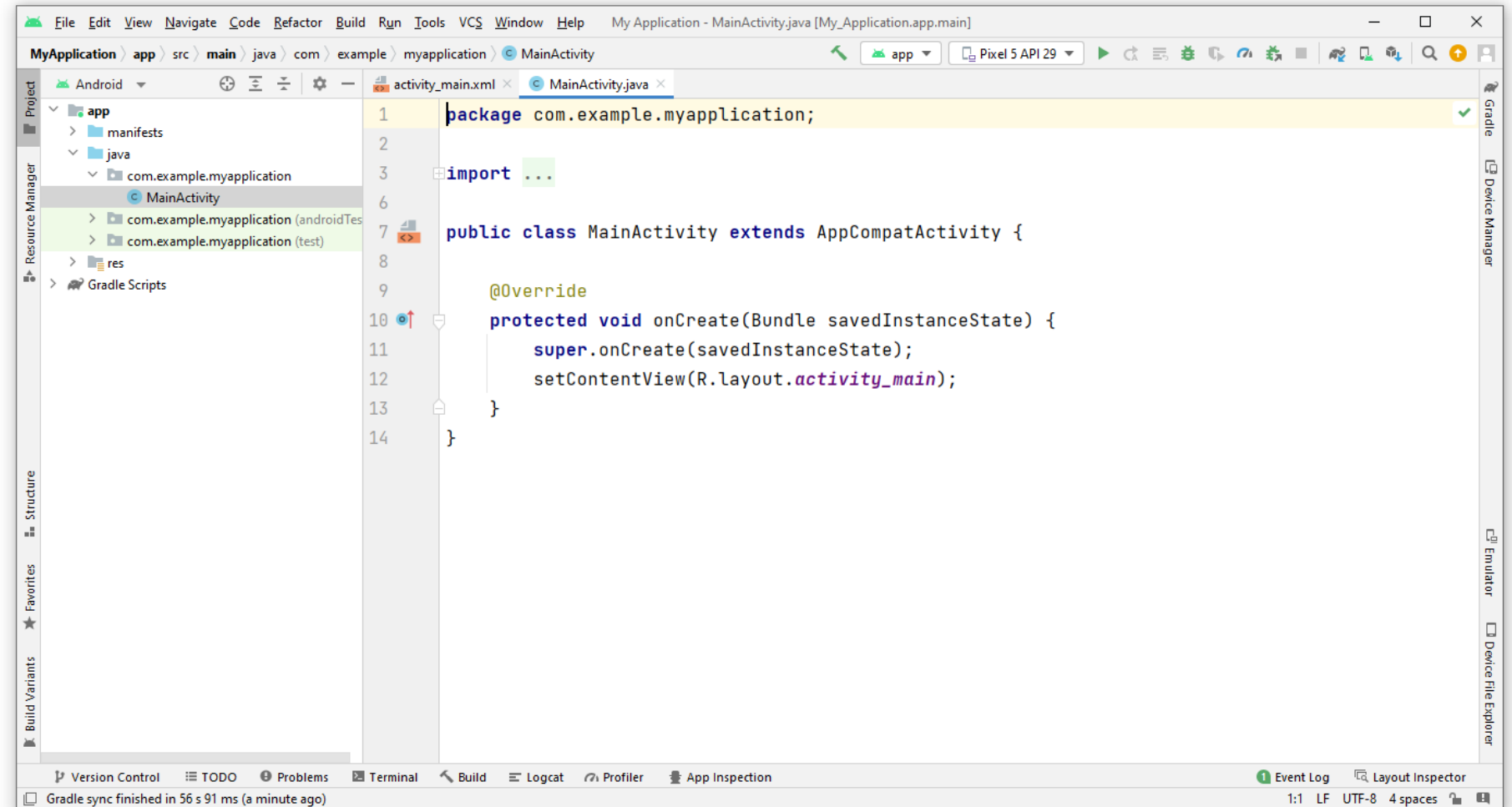
- Development Environments
- GUI Builders
- Unit Testing Tools
- Profiling, Performance Analysis, and Slicing Tools

Development Environments

- Development Environments (DE) or Integrated Development Environment (IDE)
 - Code editing
 - Compilation
 - Error detection
 - Integration with source code
 - Build/test/debugging tools
 - ...
- For example:
 - IntelliJ, Android Studio, etc.

Development Environments

- Code editing
- Compilation
- Error detection
- Integration with source code
- Build/test/debugging tools

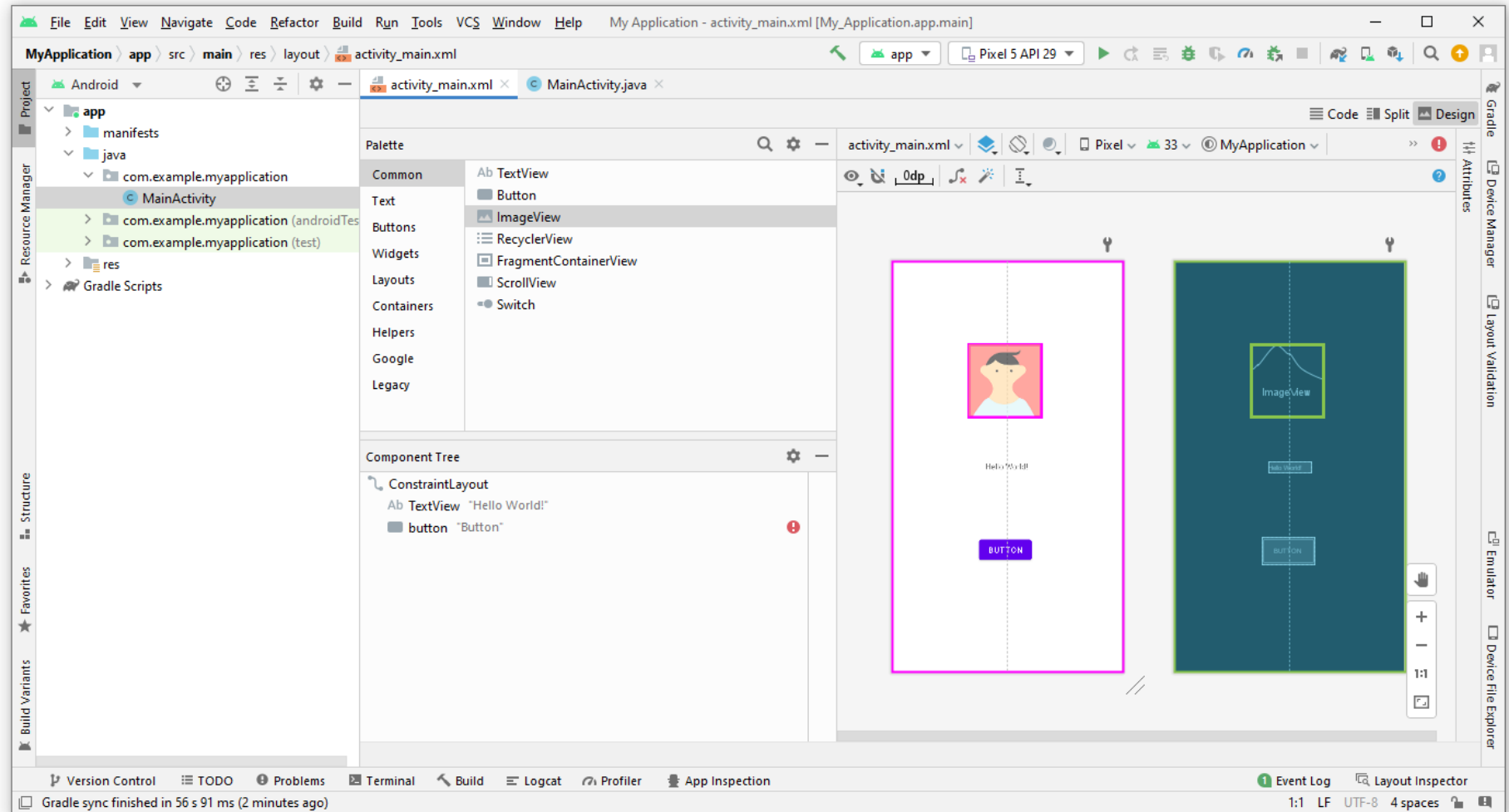


GUI Builders

- Allow you to maintain GUIs in a WYSIWYG mode
 - Visual editor
 - Drag-drop
 - Parameter setting
 - Automatic generation of source code (sometimes)
 - Code generation for event-driven development (sometimes)
 - Stand alone or integrated into an IDE
- For example:
 - Visual Studio, Android Studio, etc.

GUI Builders

- Visual editor
- Drag-drop
- Parameter setting
- Automatic generation of source code (sometimes)
- Code generation for event-driven development (sometimes)
- Stand alone or integrated into an IDE



Unit Testing Tools

- Allow you to extend and create automated testing environments
- For example:
 - JUnit, some IDEs

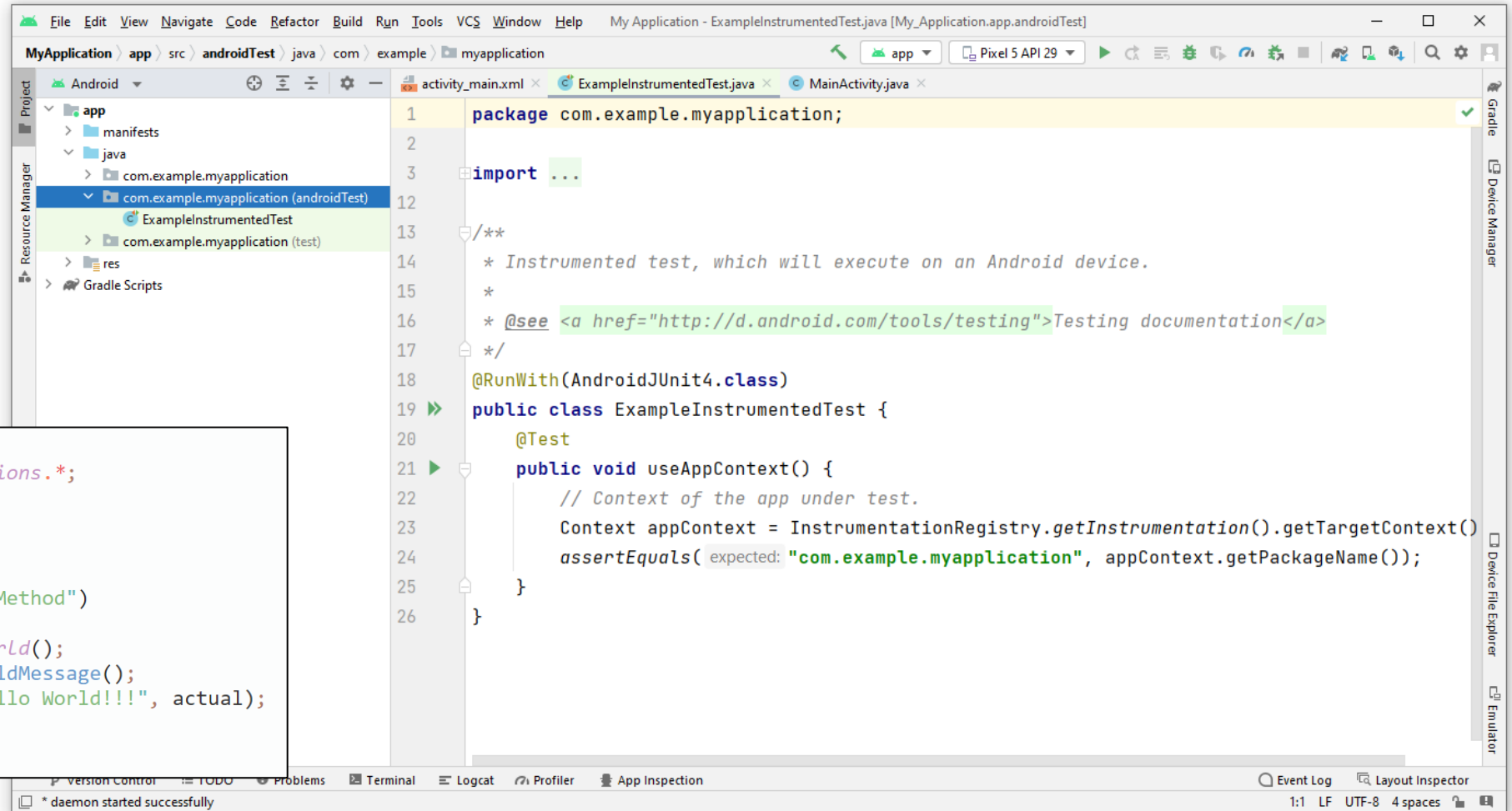
Unit Testing Tools

- Allow you to extend and create automated testing environments

```
import org.junit.jupiter.api.*;
import org.junit.jupiter.api.Assertions.*;

class HelloWorldTest {

    @Test
    @DisplayName("Test Hello World Method")
    void helloWorldTest() {
        HelloWorld hw = new HelloWorld();
        String actual = hw.helloWorldMessage();
        Assertions.assertEquals("Hello World!!!", actual);
    }
}
```



Profiling, Performance Analysis, and Slicing Tools

- Allow you to monitor your code while it runs
 - Records how many times each statement is executed
 - Records how much time is spent on each statement or path
- For example:
 - https://en.wikipedia.org/wiki/List_of_performance_analysis_tools

Product Backlog and Sprint Example

Remember: Scrum

- Scrum is a lightweight, iterative and incremental framework for managing complex work.
 - Product Backlog (initial planning)
 - Sprints (planning, design, development, testing)

Remember: Product backlog

- Product backlog
 - A breakdown of the work that needs to be done
 - Product requirements
 - I.e. user stories, use cases
 - Define features, bug fixes, non-functional requirements
 - Controlled by the “product owner”

Remember: Sprint backlog

- Sprint backlog
 - The work to be addressed during the next sprint
 - Derived from items in the product backlog
 - Broken down into tasks
 - Team members “sign up” to complete a task
 - The full sprint backlog is not always completed in one sprint

SRS Excerpt

1 Introduction

1.1 Purpose

The purpose of this document is to specify the software requirements of a software system called the New Zealand Pandemic Response Unit Tracker (NZPRU Tracker). This document will explain the capabilities of the software, its interface requirements, and both its functional and nonfunctional requirements.

1.2 Project Scope

The NZPRU Tracker is a software-intensive project to assist in containing epidemic outbreaks. During high-risk periods, people's visits to pharmacies, cafes, restaurants, supermarkets and other venues needs to be tracked. By doing so, if a person is suspected to be infected, NZPRU will be able to immediately identify all others who might have been infected by visiting the same venue at the same time.

NZPRU plans on deploying off-the-shelf tablets leveraging the venues' WiFi connectivity for collecting the remote data. Visitors to venues will be using the tablets to input their data when they enter as well as inputting the time they left on their way out of the venue

2 Overall Description

2.1 User Classes and Characteristics

NZPRU has three types of users.

- Community users: Community users do not need to login. Community users can use the mobile application to specify when they entered and exited a venue.
- Venue users: Venue users can login to the mobile application. They will login at the start of each day and logout at the end of each day. This will give the community users access to the application.
- NZPRU System administrators: Administrators can login to the web-based front end. They can view statistics, manage users.

SDS Excerpt

This will be first page shown when the application opens. In this page it will allow the installer to login with their username and password.

After he/she has login, it will allow the installer to enter the venue's details such as name and address. When this detail has been submitted it will jump to the welcome page

User ID:

Password:

Login

Information Of Venue

Name:

Address:

City:

Submit

SDS Excerpt

[Logout](#)

Welcome To
[venue Name]

Enter

Exit

Change Details

 English

 Maori

 Chinese

 Spanish

 French

 Korean

User ID: 123456

Password:

Logout

The installer is able to logout the venue through the logout button at top right corner. The logout page will provide the installer's id number and only require the installer to enter their password to logout. The logout is use for venues that will no longer needs this tablet/software.

SDS Excerpt

Are you visiting as

Individual ☒

Group ☐

Submit

For individuals, they will enter their personal details in this page such as name, email, phone number and address.

For the group, it will require one person from the group to enter their personal details and select how many people are in the group that came together on the day. For each group member they will only need to enter their first and last name. This will benefit the user without needing to line up outside for a long time waiting to enter their details in and also to help children or elderly who does not have access to phone number or email address.

SDS Excerpt

Individual

First Name:

Last Name:

Phone:

Email:

Address:

City:

Group

First Name:

Last Name:

Phone:

Email:

Address:

City:

Group Size:

1. First Name:

Last Name:

2. First Name:

Last Name:

The user will not need to enter the time and date that they have entered the venue as the system will automatically get these details and store it in database with the users' details, by doing this can save time for user when entering their details.

After the user has submitted their details it will jump to a new page with a given id number, this number will be used for reporting their exit of the venue when they leave. This number will also be send to the user's phone number and email address that they have provided in the previous page, as it can prevent the user of forgetting this number.

SDS Excerpt

ID Number

0001

*Please keep this number for reporting exit

Close

This page will automatically be closed in 20 seconds in case if the user has forgotten to close this page. When this page is closed it will go back to the welcome page.

SDS Excerpt

ID Number ☒	Please select the name that you want to exit ☐ Select All ☒ Person 1 ☒ Person 2 ☐ Person 3
Phone Number ☐	
Submit	Submit

The user will not need to enter the time and date that they leave the venue as the system will automatically get these details and store it in database with the users' details.

After they have reported their exit, it will jump to a thankyou page and this page will automatically be closed in 20 seconds in case if the user has forgotten to close this page. When this page is closed it will go back to the welcome page.

Remember: Product backlog

- Product backlog
 - A breakdown of the work that needs to be done
 - Product requirements
 - I.e. user stories, use cases
 - Define features, bug fixes, non-functional requirements
 - Controlled by the “product owner”

Product backlog example

- What are our features?
 - MoSCoW
1. Venue user tablet access
 2. Community user tablet interaction
 3. Back-end server data update
 4. ...

Product backlog example

1. Venue user tablet access

Sally Sallyson owns a café in Hamilton. Her café is only small, but is in the CBD and has great Wi-Fi coverage. She is keen to get involved in the NZPRU Tracking initiative, so she has volunteered to be one of the first venues to use the Mobile application. As a Venue User, Sally would need to be able to login and logout of the mobile application.

Frank Frankson owns a restaurant in rural Waikato. His restaurant is only small, but it is very popular with the locals. Unfortunately, the rural location means that Frank's internet connection is unstable. He is keen to be involved in the NZPRU Tracking initiative but he's not sure if his location is appropriate. As a Venue User, Frank would need to be able to login and logout of the mobile application.

Betty Bettyson owns a plant shop in Hamilton. Her plant shop is quite big, and it has a front and a back entrance. She is keen to get involved in the NZPRU Tracking initiative, so she has volunteered to be one of the first venues to use the Mobile application. She will need a tablet at each entrance to her shop. As a Venue User, Betty would need to be able to login and logout of the mobile application.

Product backlog example

2. Community user tablet interaction

Sarah Sarahson goes out to her local café at the same time every morning to get a cup of coffee. She is happy to enter her information into the NZPRU tracker, but she's always in a rush so the faster and easier it can be, the better. As a community user, Sarah would need to be able to register her entry and exit from the venue.

Harry Harryson is going out to a restaurant with his daddy-and-me group. There are four adults and seven children who join them every week. They are happy to enter their information into the NZPRU tracker, but do not want to enter information for all of the children individually. As a community user, Harry and his group would need to be able to register their entry and exit from the venue.

... one that is not happy to share personal information?

Remember: Sprint backlog

- Sprint backlog
 - The work to be addressed during the next sprint
 - Derived from items in the product backlog
 - Broken down into tasks
 - Team members “sign up” to complete a task
 - The full sprint backlog is not always completed in one sprint

Sprint backlog example

Venue user tablet access

Sally Sallyson owns a café in Hamilton. Her café is only small, but is in the CBD and has great Wi-Fi coverage. She is keen to get involved in the NZPRU Tracking initiative, so she has volunteered to be one of the first venues to use the Mobile application. As a Venue User, Sally would need to be able to login and logout of the mobile application.

1. (Development) Venue user login on tablet
2. (Development) Venue user login validation on back-end
3. (Development) Venue user logout on tablet

Sprint backlog example

Venue user tablet access

Frank Frankson owns a restaurant in rural Waikato. His restaurant is only small, but it is very popular with the locals. Unfortunately, the rural location means that Frank's internet connection is unstable. He is keen to be involved in the NZPRU Tracking initiative but he's not sure if his location is appropriate. As a Venue User, Frank would need to be able to login and logout of the mobile application.

1. (Development) Venue user login on tablet
2. (Development) Venue user login validation on back-end
3. (Development) Venue user logout on tablet
4. (Requirement/Design) Decision on supporting unstable internet (caching etc.)

Sprint backlog example

Venue user tablet access

Betty Bettyson owns a plant shop in Hamilton. Her plant shop is quite big, and it has a front and a back entrance. She is keen to get involved in the NZPRU Tracking initiative, so she has volunteered to be one of the first venues to use the Mobile application. She will need a tablet at each entrance to her shop. As a Venue User, Betty would need to be able to login and logout of the mobile application.

1. (Development) Venue user login on tablet
2. (Development) Venue user login validation on back-end
3. (Development) Venue user logout on tablet
4. (Development) Allow multiple logins at once (back-end)

Sprint backlog example

Community user tablet interaction

Sarah Sarahson goes out to her local café at the same time every morning to get a cup of coffee. She is happy to enter her information into the NZPRU tracker, but she's always in a rush so the faster and easier it can be, the better. As a community user, Sarah would need to be able to register her entry and exit from the venue.

1. (Development) Register individual entry into venue
2. (Development) Register individual exit out of venue
3. (Requirement/Design) Simple UI, Fast processing

Sprint backlog example

Community user tablet interaction

Harry Harryson is going out to a restaurant with his daddy-and-me group. There are four adults and seven children who join them every week. They are happy to enter their information into the NZPRU tracker, but do not want to enter information for all of the children individually. As a community user, Harry and his group would need to be able to register their entry and exit from the venue.

1. (Development) Register group entry into venue
2. (Development) Register group exit out of venue
3. (Development) Register children entry and exit

Putting it all together

- At the start:
 - Product Backlog for all (or MVP) features/requirements
- Next:
 - Sprint #1
 - Select items from product backlog
 - Break them down into tasks
 - Work through them until your sprint is finished
 - Sprint #2
 - Carry over any unfinished tasks from the last sprint
 - Select more items from the product backlog
 - Break them down into tasks
 - Work through them until your sprint is finished
 - Sprint #3
 - ...

At the start: product backlog

1. Venue user tablet access

Sally Sallyson owns a café in Hamilton. Her café is only small, but is in the CBD and has great Wi-Fi coverage. She is keen to get involved in the NZPRU Tracking initiative, so she has volunteered to be one of the first venues to use the Mobile application. As a Venue User, Sally would need to be able to login and logout of the mobile application.

Frank Frankson owns a restaurant in rural Waikato. His restaurant is only small, but it is very popular with the locals. Unfortunately, the rural location means that Frank's internet connection is unstable. He is keen to be involved in the NZPRU Tracking initiative but he's not sure if his location is appropriate. As a Venue User, Frank would need to be able to login and logout of the mobile application.

Betty Bettyson owns a plant shop in Hamilton. Her plant shop is quite big, and it has a front and a back entrance. She is keen to get involved in the NZPRU Tracking initiative, so she has volunteered to be one of the first venues to use the Mobile application. She will need a tablet at each entrance to her shop. As a Venue User, Betty would need to be able to login and logout of the mobile application.

2. Community user tablet interaction

Sarah Sarahson goes out to her local café at the same time every morning to get a cup of coffee. She is happy to enter her information into the NZPRU tracker, but she's always in a rush so the faster and easier it can be, the better. As a community user, Sarah would need to be able to register her entry and exit from the venue.

Harry Harryson is going out to a restaurant with his daddy-and-me group. There are four adults and seven children who join them every week. They are happy to enter their information into the NZPRU tracker, but do not want to enter information for all of the children individually. As a community user, Harry and his group would need to be able to register their entry and exit from the venue.

3. Next Feature/Requirement

...

Next: Sprint 1

Stories selected from Product Backlog:

Sally Sallyson owns a café in Hamilton. Her café is only small, but is in the CBD and has great Wi-Fi coverage. She is keen to get involved in the NZPRU Tracking initiative, so she has volunteered to be one of the first venues to use the Mobile application. As a Venue User, Sally would need to be able to login and logout of the mobile application.

Frank Frankson owns a restaurant in rural Waikato. His restaurant is only small, but it is very popular with the locals. Unfortunately, the rural location means that Frank's internet connection is unstable. He is keen to be involved in the NZPRU Tracking initiative but he's not sure if his location is appropriate. As a Venue User, Frank would need to be able to login and logout of the mobile application.

Betty Bettyson owns a plant shop in Hamilton. Her plant shop is quite big, and it has a front and a back entrance. She is keen to get involved in the NZPRU Tracking initiative, so she has volunteered to be one of the first venues to use the Mobile application. She will need a tablet at each entrance to her shop. As a Venue User, Betty would need to be able to login and logout of the mobile application.

Sprint Backlog:

1. (Development) Venue user login on tablet
2. (Development) Venue user login validation on back-end
3. (Development) Venue user logout on tablet
4. (Requirement/Design) Decision on supporting unstable internet (caching etc.)
5. (Development) Allow multiple logins at once (back-end)

Next: Sprint 2

Stories selected from Product Backlog:

Sarah Sarahson goes out to her local café at the same time every morning to get a cup of coffee. She is happy to enter her information into the NZPRU tracker, but she's always in a rush so the faster and easier it can be, the better. As a community user, Sarah would need to be able to register her entry and exit from the venue.

Harry Harryson is going out to a restaurant with his daddy-and-me group. There are four adults and seven children who join them every week. They are happy to enter their information into the NZPRU tracker, but do not want to enter information for all of the children individually. As a community user, Harry and his group would need to be able to register their entry and exit from the venue.

Sprint Backlog:

1. (Carried over) (Development) Allow multiple logins at once (back-end)
2. (Development) Register individual entry into venue
3. (Development) Register individual exit out of venue
4. (Requirement/Design) Simple UI, Fast processing
5. (Development) Register group entry into venue
6. (Development) Register group exit out of venue
7. (Development) Register children entry and exit

Next: Sprint 3

Stories selected from Product Backlog:

...

Sprint Backlog:

1. (Carried over) (Development) Register group entry into venue
2. (Carried over)(Development) Register group exit out of venue
3. (Carried over) (Development) Register children entry and exit
4. ...

COMPX₃₄₁ - Advanced Software Engineering Design



THE UNIVERSITY OF
WAIKATO
Te Whare Wānanga o Waikato